MEDIUM · ALGEBRA · ANSWER KEY

Linear equations in two variables

04

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 C

C. $y = -x - 8$

Choice C is correct. An equation of a line can be written in the form $y = mx + b$, where m is the slope of the line and $0, b$ is the y-intercept of the line. The line shown passes through the point $0, -8$, so $b = -8$. The line shown also passes through the point $-8, 0$. The slope, m, of a line passing through two points x_1, y_1 and x_2, y_2 can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. For the points $0, -8$ and $-8, 0$, this gives $m = \frac{-8 - 0}{0 - -8}$, or $m = -1$. Substituting -8 for b and -1 for m in $y = mx + b$ yields $y = -1x + -8$, or $y = -x - 8$. Therefore, an equation of the graph shown is $y = -x - 8$.

Choice A is incorrect. This is an equation of a line with a slope of -2, not -1.

Choice B is incorrect. This is an equation of a line with a slope of 1, not -1.

Choice D is incorrect. This is an equation of a line with a slope of 2, not -1.

Q2 B

B. $\frac{4}{9}$

Choice B is correct. It's given that line s is parallel to line r in the xy-plane. This means that the slope of line r is equal to the slope of line s. Line r is defined by the equation $4x - 9y = 3$. This equation can be written in slope-intercept form $y = mx + b$, where m represents the slope of the line and b represents the y-coordinate of the y-intercept of the line. Subtracting $4x$ from both sides of the equation $4x - 9y = 3$ yields $-9y = -4x + 3$. Dividing both sides of this equation by -9 yields $y = \frac{4}{9}x - \frac{1}{3}$. Therefore, the slope of line r is $\frac{4}{9}$. Since parallel lines have equal slopes, the slope of line s is also $\frac{4}{9}$.

Choice A is incorrect. This is the reciprocal of the slope of line s, not the slope of line s.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

46

The correct answer is 46. It's given that a chemist combines water and acetic acid to make a mixture with a volume of 56 milliliters (mL) and that the volume of acetic acid in the mixture is 10 mL. Let x represent the volume of water, in mL, in the mixture. The equation $x + 10 = 56$ represents this situation. Subtracting 10 from both sides of this equation yields $x = 46$. Therefore, the volume of water, in mL, in the mixture is 46.

Q4

.3846

Accept also: 5/13

The correct answer is $\frac{5}{13}$. The graph of a line in the xy-plane can be represented by the equation $y = mx + b$, where m is the slope of the line and b is the y-coordinate of the y-intercept. The given equation can be written as $y = \frac{5}{13}x - 23$. Therefore, the slope of the graph of this equation in the xy-plane is $\frac{5}{13}$. Note that 5/13, .3846, 0.385, and 0.384 are examples of ways to enter a correct answer.

Q5

C

C. $P = \frac{10}{3}s - 2$

Choice C is correct. The relationship between s and P can be modeled by a linear equation of the form P = ks + a, where k and a are constants. The table shows that P increases by 10 when s increases by 3, so k = $\frac{10}{3}$. To solve for a, substitute one of the given pairs of values for s and P: when s = 3, P = 8, so $8 = \frac{10}{3}(3) + a$, which yields a = –2. The solution is therefore $P = \frac{10}{3}s - 2$.

Choice A is incorrect. When s = 3, P = 8, but 3(3) + 10 = 19 ≠ 8. Choice B is incorrect. This may result from using the first number given for P in the table as the constant term a in the linear equation $P = ks + a$, which is true only when s = 0. Choice D is incorrect and may result from using the reciprocal of the slope of the line.

Q6

D

D. $y = 8x + 9$

Choice D is correct. An equation defining a line in the xy-plane can be written in the form $y = mx + b$, where m represents the slope and $0, b$ represents the y-intercept of the line. It's given that line t passes through the point $0, 9$; therefore, $b = 9$. The slope, m, of a line can be found using any two points on the line, x_1, y_1 and x_2, y_2, and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $0, 9$ and $1, 17$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{17 - 9}{1 - 0}$, or $m = 8$. Substituting 8 for m and 9 for b in the equation $y = mx + b$ yields $y = 8x + 9$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7

C

C. The total number of cups of jam in the large containers

Choice C is correct. It's given that the equation $4x + 6y = 36$ represents the situation where Lily filled x small containers and y large containers with all the jam she made, which was 36 cups. Therefore, $6y$ represents the total number of cups of jam in the large containers.

Choice A is incorrect. The number of large containers Lily filled is represented by y, not $6y$.

Choice B is incorrect. The number of small containers Lily filled is represented by x, not $6y$.

Choice D is incorrect. The total number of cups of jam in the small containers is represented by $4x$, not $6y$.

Q8

14.66

Accept also: `14.67`, `44/3`

The correct answer is $\frac{44}{3}$. A linear equation can be written in the form $y = mx + b$, where m is the slope of the graph of the equation in the xy-plane and $0, b$ is the y-intercept. Distributing the $\frac{1}{3}$ in the equation $y = \frac{1}{3}29x + 10 + 5x$ yields $y = \frac{29}{3}x + \frac{10}{3} + 5x$. Combining like terms on the right-hand side of this equation yields $y = \frac{44}{3}x + \frac{10}{3}$. This equation is in the form $y = mx + b$, where $m = \frac{44}{3}$ and $b = \frac{10}{3}$. Therefore, the slope of the graph of the given equation in the xy-plane is $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

Q9 D

D. $y = 4x - 5$

Choice D is correct. An equation defining a line in the xy-plane can be written in the form $y = mx + b$, where m represents the slope and $0, b$ represents the y-intercept of the line. It's given that line k passes through the point $0, -5$; therefore, $b = -5$. The slope, m, of a line can be found using any two points on the line, x_1, y_1 and x_2, y_2, and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $0, -5$ and $1, -1$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{-1 - -5}{1 - 0}$, or $m = 4$. Substituting 4 for m and -5 for b in the equation $y = mx + b$ yields $y = 4x - 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10 1.636

Accept also: 18/11

The correct answer is $\frac{18}{11}$. For a line that passes through the points x_1, y_1 and x_2, y_2 in the xy-plane, the slope of the line can be calculated using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that a line passes through the points $4, 6$ and $15, 24$ in the xy-plane. Substituting $4, 6$ for x_1, y_1 and $15, 24$ for x_2, y_2 in the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$, yields $m = \frac{24 - 6}{15 - 4}$, or $m = \frac{18}{11}$. Therefore, the slope of the line is $\frac{18}{11}$. Note that 18/11 and 1.636 are examples of ways to enter a correct answer.

Notes

Accept equivalent forms (e.g. `0.25` ≡ 1/4). For grid-in items, decimal and fraction answers are both valid.

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